Carbohydrate catabolism: glycolysis and fermentation

D-glucose is one of the major sources of **energy** used by organisms, occupying a central position in our metabolism. Glucose is particularly important for some tissues and cell types; erythrocytes, for example, depend almost exclusively on glucose for ATP production. The free energy change of the complete oxidation of glucose to CO_2 and H_2O is -2840 kJ/mol.

Glucose is also used as a source of **precursors** for biosynthetic reactions.

The main routes of glucose utilization are four:

- **Storage**: most glucose is stored in the form of a polymer of high molecular mass: starch (plants) or glycogen (animals). In this way, the cell stores an important source of energy which, in situations of need, will allow a rapid supply of energy.

- Oxidation via **glycolysis**: to a three-carbon compound, pyruvate.

- Oxidation via the **pentose phosphate pathway**: for the formation of NADPH (necessary for biosynthesis reactions) and ribose 5-phosphate (for the synthesis, among others, of RNA, DNA and nucleotide coenzymes).

- **Synthesis**: e.g. of structural polymers, or of precursors for the synthesis of other molecules such as amino acids, lipids, etc.

Glycolysis is the central route of glucose catabolism. Glycolysis is defined as the **sequence of enzymatic reactions that convert glucose to pyruvate, in a pathway that does not require O_2, with production of ATP and NADH**.

Characteristics of glycolysis

1. Glycolysis was one of the first metabolic pathways to originate in evolution. It is therefore an almost **universal** pathway, differing only from one species to another in the details of its regulation and in the fate of the newly formed pyruvate; whereas the reactions that take place, as well as the enzymes, are practically the same.

2. It is composed of **10 enzymatically catalyzed reactions**. Three of these enzymes constitute the major regulatory points of the pathway.

3. It is **amphibolic**, that is, it has both catabolic (generation of energy) and anabolic (generation of precursors) functions.

4. It takes place in the **cytoplasm**.

5. Although most individual reactions are reversible, overall, it is highly exergonic and proceeds essentially **irreversibly** in the glycolytic direction.

6. It can occur under **anaerobic conditions**.

7. All glycolysis intermediates are **phosphorylated**.

Phosphorylation of metabolites has several functions:

- At physiological pH the phosphate group is negatively charged, thus preventing phosphorylated metabolites from diffusing out of the cell (the plasma membrane is impermeable to polar compounds). Thus, despite the large difference in extra- and intracellular concentrations of these compounds, the cell does not need to expend energy to retain them.

- Phosphorylation increases the free-energy content of the metabolite, thereby "energizing" the molecule and facilitating subsequent metabolic transformations. Some phosphorylated intermediates have particularly high phosphoryl-group transfer potentials, which can be used to drive ATP synthesis.

Phases of glycolysis

Glycolysis is divided into two phases:

- **Preparatory phase**: consisting of the first five steps of the pathway (up to the formation of glyceraldehyde 3-phosphate). In this phase, **energy is invested** from ATP, raising the free energy content of the metabolic intermediates. The other sugars are incorporated into the pathway at some point during this phase.

- **Energy-producing phase**: consisting of the last five reactions, up to the formation of pyruvate. **Energy is obtained** in the form of ATP and reducing power (NADH).

Reactions of glycolysis

1. Phosphorylation of glucose

The first reaction of glycolysis is the phosphorylation of glucose to give **glucose 6-phosphate**. Under cellular conditions the reaction is highly exergonic and irreversible. When glucose enters the cell, it is rapidly phosphorylated, so that the concentration of free glucose in the cell is very low. By the phosphorylation reaction at carbon 6 the glucose will be activated for subsequent reactions. The donor of the phosphate group is ATP.

This reaction is catalyzed by ***hexokinase***. There are 4 hexokinase isoenzymes (I to IV). In muscle, I, II and III act mainly and have the following characteristics:

1- K_m (glucose) = 0.1 mM (normal blood glucose concentration is 4-8 mM).

2- Low specificity. It can phosphorylate other hexoses such as fructose or mannose.

3- Its activity is allosterically inhibited by its own product, glucose 6-phosphate.

In the hepatic cells and pancreatic β cells there is another isoform of the enzyme called ***glucokinase*** (or hexokinase IV). Its characteristics are the following:

1- It has a lower affinity for glucose than hexokinase: K_m (glucose) = 5-10 mM. Therefore, in order to act it needs higher concentrations of glucose.

2- It is more specific, acting mainly on glucose.

3- Unlike hexokinase, glucokinase is not inhibited by glucose 6-phosphate. In hepatocytes, fructose 6-phosphate promotes the binding of glucokinase to glucokinase regulatory protein (GKRP), resulting in inhibition of the enzyme.

Therefore, **glucose phosphorylation in the liver depends on the abundance of glucose and not on the need for its products**. Moreover, glucose will only be utilized in the liver when there are high concentrations of this metabolite.

2. Isomerization to fructose 6-phosphate

It is the change from an aldose to a ketose, forming **fructose 6-phosphate**. This reaction is catalyzed by ***phosphoglucose isomerase*** (or phosphohexose isomerase). It is a reversible reaction (it proceeds in one direction or the other depending on the concentration of the metabolites).

3. Phosphorylation of the fructose 6-phosphate

Fructose 1,6-bisphosphate is produced in a reaction catalyzed by ***phosphofructokinase-1*** (PFK-1) in which ATP is required as a donor of the phosphate group. This is the critical control point for the progression of glycolysis, so the activity of this enzyme is controlled allosterically in a very precise manner. In general, the activity increases when the energy load of the cell decreases and decreases when the energy load increases. Under physiological conditions the reaction is irreversible.

4. Cleavage of fructose 1,6-bisphosphate

A reaction catalyzed by ***aldolase*** in which fructose 1,6-bisphosphate is cleaved into two three-carbon molecules: **dihydroxyacetone phosphate** and **glyceraldehyde 3-phosphate**. Under physiological conditions it is a reversible reaction. The name of the enzyme derives from the nature of the reverse reaction, the aldol condensation.

5. Interconversion of triose phosphates

Glyceraldehyde 3-phosphate is found in the direct pathway of glycolysis, but not dihydroxyacetone phosphate. The latter can be readily converted to **glyceraldehyde 3-phosphate** by reversible isomerization catalyzed by ***triose phosphate isomerase***.

At this point the preparatory phase of glycolysis ends. It is important to note that from one molecule of glucose 2 molecules of glyceraldehyde 3-phosphate have been obtained. Both molecules enter the energy-producing phase.

6. Oxidation of glyceraldehyde 3-phosphate

This is a reversible reaction catalyzed by ***glyceraldehyde-3-phosphate dehydrogenase*** (**GAPDH**) in which oxidation and phosphorylation occur to yield **1,3-bisphosphoglycerate**. In the oxidation reaction, an anhydride group called acylphosphate, a high-energy phosphate compound, is

produced. On the other hand, electrons from the oxidation of carbon 1 (from carbonyl group to carboxyl group) reduce an NAD^+ to **NADH**.

There are chemical compounds that affect the function of GAPDH. Arsenate (AsO_4^{3-}) is a structural analog of phosphate that is used by GAPDH to give 1-arseno-3-phosphoglycerate, an unstable compound that spontaneously hydrolyzes to 3-phosphoglycerate (preventing ATP formation in the next step of glycolysis). This is one of the mechanisms mediating arsenate toxicity.

7. ATP formation from 1,3-bisphosphoglycerate

The high phosphate potential of 1,3-BPG is used to generate **3-phosphoglycerate** and **ATP** (in a process known generically as substrate-level phosphorylation). The reaction is reversible and is catalyzed by ***phosphoglycerate kinase***.

8. Conversion of 3-phosphoglycerate to 2-phosphoglycerate

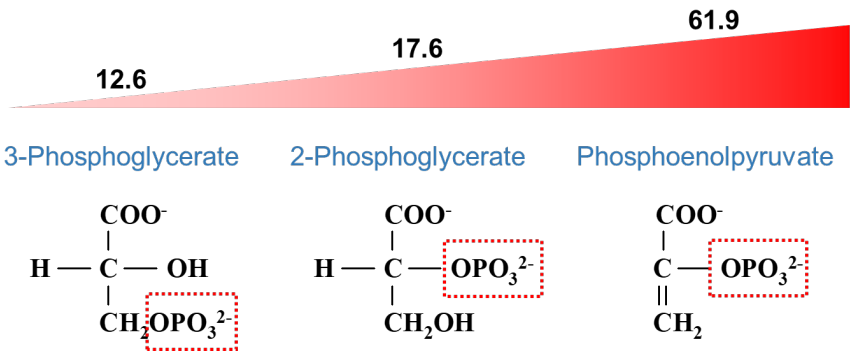
In this reaction an intramolecular rearrangement occurs, shifting the phosphate group between carbon 3 and carbon 2 to give rise to **2-phosphoglycerate**. The reaction is reversible and is catalyzed by ***phosphoglycerate mutase***. The term mutase is often applied to isomerases that catalyze the transfer of a functional group from one position to another of the same molecule. This is the first transformation aimed at increasing the phosphorylation potential of the metabolic intermediate.

9. 2-Phosphoglycerate dehydration

This is a dehydration reaction catalyzed by ***enolase*** in which **phosphoenolpyruvate**, a compound of high phosphate group transfer potential, is produced.

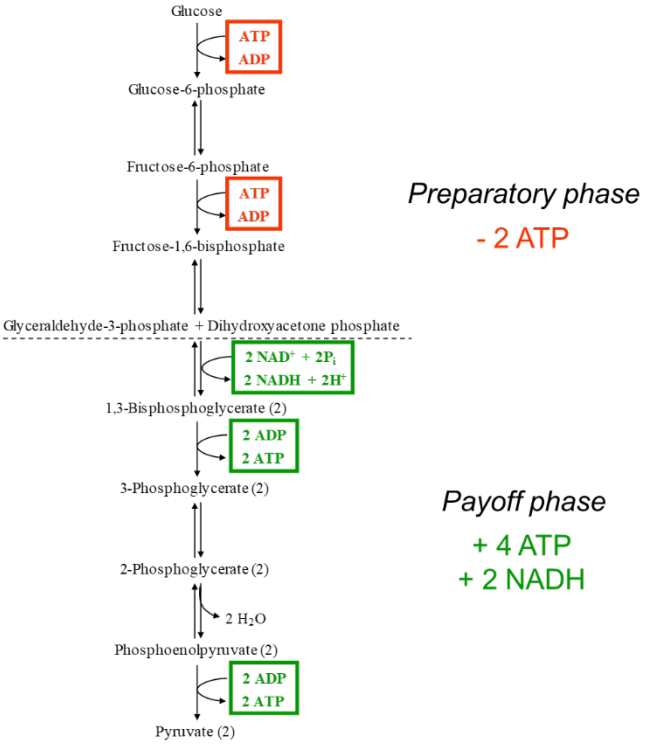
10. Pyruvate formation

Through the two previous reactions of glycolysis, the potential of the phosphate group has been increased to a level sufficient for its hydrolysis to provide enough energy for the formation of an ATP molecule.



Thus, in this last reaction of glycolysis, the transfer of the phosphate group from phosphoenolpyruvate to ADP occurs, catalyzed by ***pyruvate kinase***, giving rise to the final product of the process, **pyruvate**. This is the second substrate-level phosphorylation of glycolysis. The reaction is irreversible and is an important point in the regulation of this metabolic pathway.

Overall balance of glycolysis



The net balance of glycolysis is:

Glucose + 2 ADP + 2 P_i + 2 NAD^+ → 2 pyruvate + 2 ATP + 2 NADH + 2 H^+ + 2 H_2O

Thus, in the conversion of glucose to pyruvate, two ATP molecules are generated. Under aerobic conditions, the energy stored in NADH can also be used to generate additional ATP through oxidative phosphorylation.

Regulation of glycolysis

Glycolysis is regulated so that ATP levels and precursor supply for biosynthetic processes are adequate.

Adjustment in the rate of glycolysis is done primarily **by controlling the activity of the enzymes that catalyze the three irreversible reactions**: *hexokinase*, *phosphofructokinase-1* and *pyruvate kinase*.

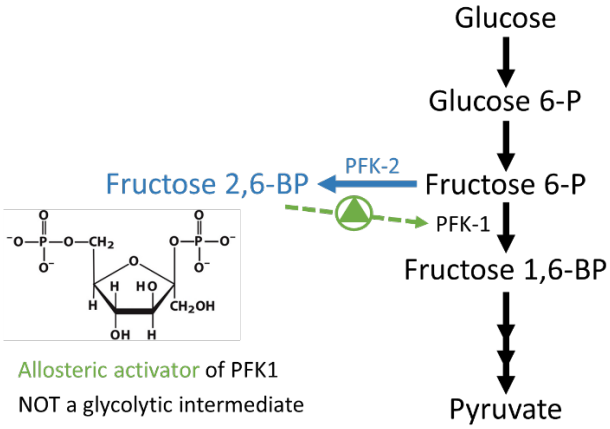
Regulation of *hexokinase*

Glucose 6-phosphate is an allosteric inhibitor of *hexokinase*. In the liver there is *hexokinase IV*, also called *glucokinase*. ***Glucokinase* is not inhibited by glucose 6-phosphate**, which allows the supply of glucose 6-phosphate for hepatic glycogen synthesis.

Regulation of *phosphofructokinase-1*

Phosphofructokinase-1 (PFK-1) contains regulatory sites to which allosteric inhibitors (such as ATP, H+ or citrate) or allosteric activators (such as AMP, ADP or fructose 2,6-bisphosphate) can bind. In general, PFK-1 is activated with low energy and few biosynthetic precursors.

Fructose 2,6-bisphosphate is a **potent activator of PFK-1**, particularly important in the liver. Fructose 2,6-bisphosphate is synthesized from fructose 6-phosphate by the action of ***phosphofructokinase-2*** (PFK-2/FBPase-2).



The concentration of fructose 2,6-bisphosphate inside cells is regulated by the blood glucose concentration through intracellular signaling induced by the hormones insulin, glucagon and epinephrine. For example, in hepatocytes, activation of the glucagon receptor, in response to low blood glucose levels, results in decreased PFK-2 activity, decreasing fructose 2,6-bisphosphate levels and thus the activity of glycolysis.

Regulation of *pyruvate kinase*

Pyruvate kinase is **allosterically inhibited by high concentrations of ATP and alanine**, and **activated by fructose 1,6-bisphosphate** through a feed-forward mechanism. In addition, the hepatic isoform can also be **inhibited by phosphorylation**. Low blood glucose levels lead to phosphorylation of the enzyme in hepatocytes through activation of the glucagon receptor.

Catabolic fates of pyruvate

Pyruvate synthesized in glycolysis can undergo two types of degradation depending on the cellular context and mitochondrial oxidative capacity:

- Pyruvate can enter mitochondria and be converted to acetyl-CoA, which can subsequently be oxidized to CO_2 and H_2O through the citric acid cycle.

- When the rate of NADH production exceeds the capacity for mitochondrial oxidation, NAD^+ can be regenerated by a process known as **fermentation**. There are many different types of fermentation, such as lactic fermentation or alcoholic fermentation.

Lactic fermentation

Pyruvate can be reduced to **lactate** by ***lactate dehydrogenase***. In this reaction, NADH is oxidized to NAD^+, allowing glycolysis and ATP production to continue when the rate of NADH production exceeds the capacity for mitochondrial oxidation. Lactate formation occurs in **erythrocytes**, which lack mitochondria, and increases markedly in **skeletal muscle during intense exercise**.

The balance of glycolysis with lactic fermentation is:

Glucose + 2 ADP + 2 P_i → 2 lactate + 2 ATP + 2 H_2O

Lactate is not simply a waste product. It can be transported through the blood and used as a metabolic fuel by other tissues or converted back to glucose in the liver through the **Cori cycle**.

During very intense muscular activity, high ATP turnover and glycolytic flux are accompanied by H^+ accumulation and increased lactate production. If proton production exceeds the buffering and removal capacity of the tissue, intracellular pH decreases. The resulting acidification inhibits PFK-1, which is sensitive to low pH, thereby reducing the rate of glycolysis and helping to limit sustained vigorous activity. If the production of H^+ exceeds its removal from the blood, blood pH can also decrease, leading to metabolic acidosis. Because this situation is typically associated with high blood lactate concentrations, it is commonly referred to as lactic acidosis, although lactate itself is not the direct cause of the fall in pH.

Other organisms also ferment glucose and other hexoses to lactate. For example, in yogurt and cheese production, certain lactobacilli and streptococci ferment milk lactose to lactate. This acidifies the medium, leading to denaturation and precipitation of milk casein, among other proteins.

Ethanol fermentation

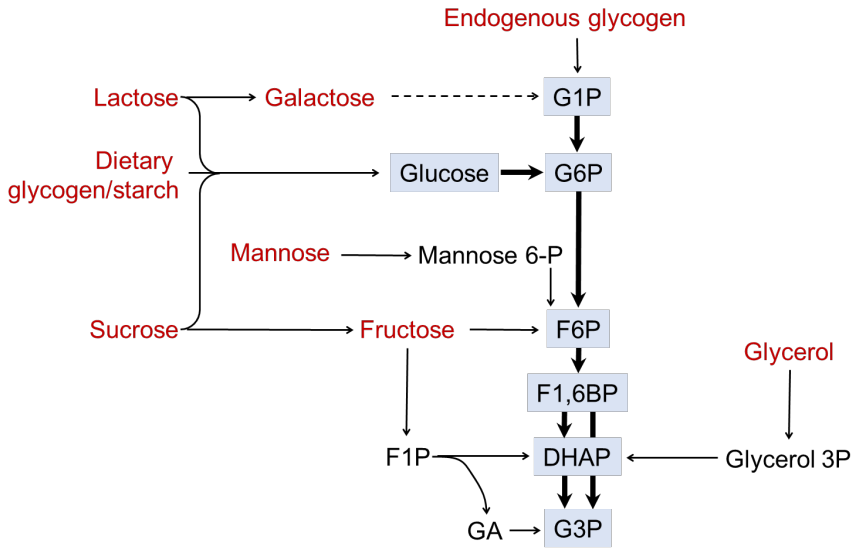
It occurs in two reactions, a decarboxylation of pyruvate to **acetaldehyde**, catalyzed by ***pyruvate decarboxylase***, and a reduction to **ethanol**, catalyzed by ***alcohol dehydrogenase***. *Pyruvate decarboxylase* is found in yeast and other microorganisms. This enzyme uses **thiamine pyrophosphate** (TPP) as a coenzyme. TPP is derived from vitamin B_1 or **thiamine**. *Pyruvate decarboxylase* is not present in human cells. *Alcohol dehydrogenase* is present, but functions mainly in the opposite direction, catalyzing the oxidation of ethanol to acetaldehyde during **metabolization of ethanol**.

Balance of alcoholic fermentation: Glucose + 2 ADP + 2 P_i → 2 ethanol + 2 ATP + 2 CO_2 + 2 H_2O

This fermentation is used in many industrial processes such as the production of bread and beer.

Entry of other sugars into glycolysis

In addition to glucose, many other carbohydrates enter the glycolytic pathway to be degraded and produce energy. The following scheme shows some examples.



Monosaccharides such as mannose are simply phosphorylated and isomerized to glycolytic intermediates; whereas fructose, which comes from fruits or sucrose, follows different degradation routes in hepatocytes and extrahepatic tissues. In the case of galactose, which comes mainly from lactose, a complex process involving several enzymes takes place. **Galactosemia** is a rare disease caused by genetic deficiency of some of the enzymes involved in galactose catabolism, mainly GALT. Accumulation of galactose-1-phosphate (e.g. after ingestion of milk) causes vomiting, diarrhea, hepatomegaly, cataracts and intellectual disability.

In the case of disaccharides, they cannot enter the cells without first being hydrolyzed to monosaccharides. The enzymes that hydrolyze them are bound to the external surface of the epithelial cells lining the small intestine. The monosaccharides are transported to the interior of the epithelial cells, from which they pass into the blood and are transported to the various tissues where they are phosphorylated and channeled into the glycolytic pathway. **Lactose intolerance** is a common condition in many adult populations. It is due to the decline in ***lactase*** activity in the intestinal cells after infancy. As a result, lactose cannot be digested or absorbed and is converted by bacteria in the large intestine into organic acids and gases leading to abdominal discomfort, bloating and diarrhea.

In the case of dietary polysaccharides, mainly starch and glycogen, salivary and pancreatic α-amylases degrade them to oligosaccharides and disaccharides. They are then degraded to monosaccharides by hydrolases in the intestinal walls.

Importance of glycolysis in cancer: The Warburg effect

Tumor cells often show a very high rate of glucose uptake and glycolysis. A substantial fraction of the pyruvate produced is converted into lactate even when oxygen is available, a phenomenon known as aerobic glycolysis or the **Warburg effect**, first described by Otto Warburg around 100 years ago.

The Warburg effect does not generally imply that tumor mitochondria are defective. Instead, increased glycolytic flux can provide ATP at a high rate and, importantly, metabolic intermediates that can be used for the synthesis of nucleotides, amino acids and lipids required for cell proliferation. Increased glycolytic activity and the associated export of lactate and protons contribute to acidification of the tumor microenvironment, which can influence tumor progression and response to therapy. **The high glucose uptake of many tumors has an important clinical application**. It allows their detection by positron emission tomography (PET) after administration of the glucose analog ^{18}F-fluoro-2-deoxyglucose (^{18}F-FDG), which is preferentially taken up by highly glycolytic tissues. After entering the cell, ^{18}F-FDG is phosphorylated but cannot proceed normally through glycolysis, resulting in its intracellular accumulation.

Author: Ruben Pio

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